



Q1. What are the most commonly-available IC packages? What are DIP, SMD, QFP and BGA IC packages?

Anand

A1. An integrated circuit, or IC, package is a case that surrounds the circuit or semiconductor device. The package is used to protect circuit materials from corrosion and physical damage, and allows easy handling and mounting of electrical contacts while connecting these to the PCB.

Many modern ICs including microprocessors, microcontrollers (MCUs) and sensors come in various packages. For example, the popular ATmega328 MCU used in Arduino comes in both dual-in-line package (DIP) and thin-quad-flat package (TQFP), while an accelerometer or a gyroscope like MPU-6050 comes in quad-flat no-leads (QFN) form.

There are many types of IC packages, each having unique dimensions, mounting types and/or pin counts. Most common IC package types include DIP, surface-mount device (SMD), small-outline package (SOP), quad-flat package (QFP) and ball-grid array (BGA).

Dual-in-line package (DIP). It is the most common through-hole IC package used in circuits, especially hobby projects. This IC has two parallel rows of pins extending perpendicularly out of a rectangular plastic housing.

Overall dimensions of a DIP package depend on its pin count. The most common pin counts are four, six, eight, fourteen, eighteen, twenty, twenty-eight and forty pins. The pins on a DIP IC are spaced 2.54mm apart, which is a standard spacing—perfect for fitting into breadboards, veroboards and other



Some common IC packages

prototyping boards.

A DIP IC can also be easily soldered on PCBs. Sometimes, instead of soldering an IC directly to the PCB, an IC socket is used. Using the socket allows for the DIP IC to be removed from and inserted into the PCB easily.

Surface-mount device (SMD). There are a variety of surface-mount packages including SOP, small-outline transistor (SOT) and QFP available in the market. SMD IC packages usually need custom PCBs, containing a matching pattern of copper on which they are to be soldered. Usually special automated tools are used to solder these on PCBs.

Small-outline IC (SOIC) package. SOIC package is shorter and narrower than DIP. It is an SMD with all DIP pins bent outwards and shrunk down to size. Each pin is usually spaced about 1.27mm from the next.

Small-outline package (SOP). This is an even smaller version of SOIC package. Similar to SOIC, SOP family has a smaller form factor, with pin spacing less than

1.27mm. Each SOP includes a plastic small-outline package (PSOP), thin small-outline package (TSOP) and thin-shrink small-outline package (TSSOP).

Quad-flat package (QFP). Unlike DIP having two sides, QFP IC has pins on all four sides. A QFP IC can have pins anywhere from eight per side

(32 total) to upwards of seventy (300+). Pins on a QFP IC are usually spaced anywhere from 0.4mm to 1mm apart. Smaller variants of the standard QFP package include thin QFP (TQFP), very-thin QFP (VQFP) and low-profile QFP (LQFP) packages.

Quad-flat no-leads (QFN) package. There is another QFP IC type, but with a different pin structure, called QFN package. Pins on a QFN package are exposed on the bottom and sometimes on both the sides and the bottom.

Small-outline transistor (SOT). SMD devices such as rectangular transistors are available in SOT packages.

Ball-grid array (BGA). Advanced ICs are available in BGA packages. These amazingly intricate packages have small balls of solder arranged in a 2D grid on the bottom. Usually, to put these packages onto a PCB requires an automated procedure involving pick-and-place machines and reflow ovens. BGA packages are found on pcDuino and Raspberry Pi boards.

Q2. What are the differences between SMD, SMT and THT?

Rajesh

A2. Fundamentally, there is a difference between surface-mount technology (SMT) and surface-mount device (SMD). It is because one is a process, whereas the other is a device.

SMT is the method of placing components (like an SMD) on the board. Here, a calculated amount of solder

paste is applied to the board. Then, the pick-and-place machine is used to mount the SMD component onto the board.

An SMD is an electronic component that can be found on a PCB. It is designed to be mounted directly on PCB.

Through-hole-technology (THT)

involves inserting leads of components into drilled holes in the PCB. Leads can then be soldered onto pads or lands on the solder side of the board, usually with a simple soldering iron by hand or wave soldering process. Soldering a DIP IC on the PCB is an example of THT.